

Specialized recall procedures

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Studies of memory search using the free recall paradigm have advanced our understanding of human memory for more than a century. Here, we review seven specialized recall procedures that depart from methodological orthodoxy: the inter-list repetition procedure, the list-before-last procedure, the recall-by-category procedure, the final-free recall procedure, the overt-rehearsal procedure, the externalized recall procedure, and the spatial-temporal recall procedure. Key empirical findings from each of these procedures have challenged existing theories and led to theoretical advances.

Recall and recognition constitute the two classic procedures for studying explicit, episodic memory. In both procedures, the experimenter asks subjects to remember items experienced on a recent list. Although the subject may or may not know that they will be tested on the items, the recognition and recall procedures explicitly require subjects to consult their memory for an item's prior occurrence. Students of memory possess considerable familiarity with the classic variants of recognition (e.g., item vs. associative) and recall (e.g., cued recall, serial recall, and free recall). Our focus here is on the free recall task, and in particular, seven specialized procedures that researchers have developed to better understand context-based episodic memory. Before introducing these seven procedures, we briefly describe the standard free recall procedure.

In free recall, subjects typically study a list of words, and then, either immediately or following a filled retention interval, they attempt to recall as many words as they can remember in any order. Because subjects can recall items without regard to their order of study, the statistical regularities in their recall order offer clues to the organization of memory. These regularities include the serial position of the item produced during recall initiation (i.e., the first recall response): this is typically one of the final list items when recall immediately follows study, and is typically one of the first list items when recall follows a filled delay period. Recall transitions reveal further regularities, including the tendency to

cluster recalls based on their temporal similarity (the contiguity effect), semantic similarity, spatial proximity, emotional valence, or other source attributes. Previous reviews summarize these and other free recall phenomena (Kahana, 2017, 2020; Healey, Long, & Kahana, 2019).

Standard variants of free recall include manipulations of encoding task (Polyn, Norman, & Kahana, 2009; Long & Kahana, 2017), modality of presentation (Murdock & Walker, 1969; Pazdera & Kahana, 2023), presentation rate and list length (Murdock, 1962; Ward, Tan, & Grenfell-Essam, 2010), intra-list repetition (Madigan, 1969; Melton, 1970; Siegel & Kahana, 2014), intra-list semantic similarity (Romney, Brewer, & Batchelder, 1993), and categorical organization of lists (Pollio, Richards, & Lucas, 1969; Tulving & Pearlstone, 1966; Ward & Tan, 2023), among many others (Kahana, 2012). Another standard variant involves giving subjects multiple trials to learn a list (multi-trial free recall) with list order either being constant or variable across presentations (Tulving, 1962; Klein, Addis, & Kahana, 2005). Yet another variant, continual-distractor free recall, has subjects perform distractor tasks between the encoding of different list items and also at the end of the study list. This procedure has revealed that recency and contiguity effects appear at both short- and long-time scales, arguing against classic dual-store models of free recall (Bjork & Whitten, 1974; Howard & Kahana, 1999; Neath, 1993).

The great diversity of these procedural variations obscures two critical features of free recall. The first concerns the nature of the target list, which is generally the last set of experienced items. Shared semantic attributes rarely define these items; rather, a span of time and a designation of list membership define them. A second canonical feature of free recall concerns the independence of the target list from other prior experiences. In a typical experiment, subjects study and attempt to recall multiple lists, with each list being in-

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dependent of the others (Adroque, Herz, Halpern, Tracy, & Kahana, 2024). Both of these features differ markedly from real-world uses of memory. First, we rarely face the challenge of recalling items defined solely based on their joint occurrence on a study list. Second, we often seek to recall memories that share attributes other than their belonging to a recent set of experienced items.

Below, we review seven specialized recall procedures that each deviate in some critical way from the standard procedures described above. Figure 1 depicts the first six of these procedures. 1) In the inter-list repetition procedure, study-test lists harbor highly similar or identical items, thus challenging subjects' ability to focus retrieval on the just-presented list. 2) In the list-before-last procedure, following the study of list i in a series of lists, subjects attempt to recall list $i - 1$, while suppressing recall of items from list i . This procedure breaks a key feature of standard free recall by having subjects target a set of items other than the most recent list. This change arguably makes the task more similar to everyday recall situations outside the laboratory. 3) In the recall-by-category procedure, subjects study lists composed of several exemplars from each of several taxonomic categories (e.g., birds or tools). Then, at test, the experimenter asks for free recall of all exemplars of a cued category. This procedure is a twist on standard categorized free recall in which subjects may recall all of the items, both within and across categories, during the recall period (e.g., Pollio et al., 1969). 4) In the final-free recall procedure, subjects use standard free recall procedures to study and recall a series of lists. Following the last study-test lists, the experimenter asks the subjects to recall all items from all studied lists in any order. 5) The overt-rehearsal procedure asks subjects to verbalize items that come to mind between item presentations, as they study a list for a subsequent recall test. This procedure provides researchers with valuable information on subjects' rehearsal patterns, which in turn predict subsequent recall behavior. 6) The externalized free recall procedure also asks subjects to verbalize words that come to mind, but during recall rather than during encoding. By indicating, with a keypress, which verbalizations they made in error, subjects provide valuable information on how they filter or inhibit responses during recall. 7) In hybrid spatial-temporal recall tasks, lists of items unfold in time and space, usually through a cover-task in which subjects experience items as they navigate through a virtual environment (see Figure 7A). This variant shares features of continual-distractor free recall in that subjects perform a task (navigation) between each encoded item. Whereas standard free recall places items from a high-dimensional semantic space along a one-dimensional temporal dimension, these tasks place these items within a multivariate spatiotemporal context. In this sense, space acts as a second dimension of context, which may be correlated to various degrees with time but allows for revisiting prior

contexts during a single list presentation.

The Inter-list Repetition Procedure

Murdock's classic (1962) study of the serial position effect in free recall focused the field's attention on *intra-list* recall phenomena, such as the primacy and recency effects, and on theoretical models aimed at explaining the rich set of phenomena obtained in single-trial free recall. This contrasted with earlier research in the verbal learning tradition, reviewed by Crowder (1976), that emphasized the paired-associate method and used manipulations of similarity and repetition to investigate the effects of proactive interference (PI) and retroactive interference (RI). Although early researchers attracted to the free recall paradigm tended not to focus on classic interference manipulations, it was not long before researchers applied interference ideas and inter-list manipulations to the study of free recall.

In standard free recall, there is always a single target list whose items the subject attempts to recall on a given trial. Subjects study and recall many such lists over the course of an experimental session. Words normally do not repeat across neighboring target lists, which minimizes inter-list interference effects. Nonetheless, performance generally decreases across trials (Kahana, Aggarwal, & Phan, 2018), consistent with proactive interference caused by items on prior lists interfering with memory for the target list (Underwood, 1957).

In the inter-list repetition procedure, the target lists include a mixture of items presented on prior lists (repeat items) or items semantically similar to those on prior lists, along with new items (semantically unrelated to prior list items). This repetition of semantic content can produce marked interference. Compared with standard free recall, the inter-list repetition procedure more closely mimics many real-life situations. When shopping for groceries, for example, we must remember our current needs, but in doing so we must not be overwhelmed by our memories of other recent shopping events that include both overlapping and non-overlapping items. A professor who is teaching two sections of related (or identical) courses must somehow recall what was said in the prior lecture of the target class without being overwhelmed by memories of the other section or the related class. In each of these scenarios, the memory search set often includes items experienced in similar recent contexts, thus taxing our ability to selectively target information experienced within or appropriate to a given context. The inter-list repetition procedure thus provides novel insights into how subjects can associate items with cues that support the recall of other items from the proper (but not improper) context.

In an early study of inter-list repetition effects in free recall, Anderson and Bower (1972) asked subjects to study lists with many overlapping items across lists. In one experiment, a standard immediate free recall period followed each list. In

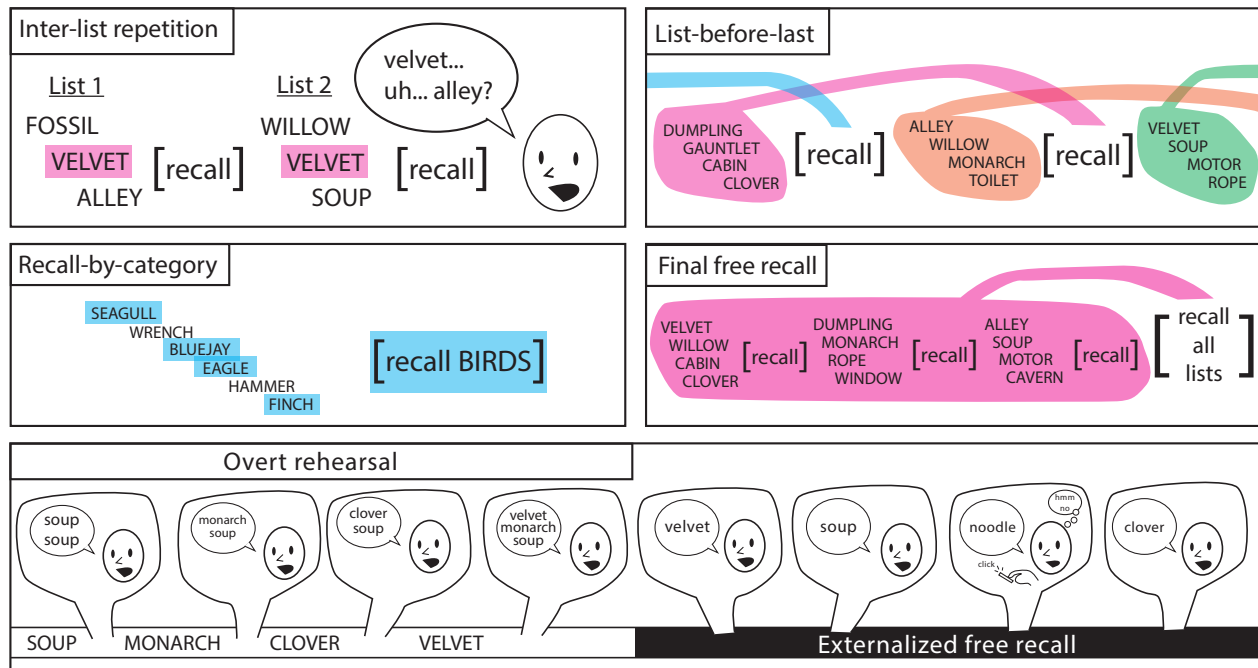


Figure 1. A schematic overview of six specialized recall tasks. Inter-list repetition: Critical items are repeated across lists, challenging a subject's ability to focus recall on the most recent list. List-before-last: Rather than recall the most recent list of study items, the subject is asked to recall the list studied preceding the most recent list. Recall-by-category: The subject is given a cue prompting them to recall a subset of the studied items. Final free recall: At the end of a session, the subject is asked to recall study items from all of the study-test lists. Overt rehearsal: The subject is encouraged to verbalize any items that come to mind as the study list progresses. Externalized free recall: The subject is encouraged to verbalize any items that come to mind during the free recall period, with a button to indicate when a particular response was not from the study list.

another experiment, subjects attempted to recall any item that had been presented up to that point in the experiment, and for each recalled item, they rated their confidence in that item belonging to the most recent list. They found that as subjects encountered more and more repetitions of items across lists, these judgments became less confident. To explain this result, they proposed that subjects encode a time-varying signal to differentiate a given word's occurrence on the target list from the same word's occurrence on other lists. Subjects use this temporal information to restrict their recall to only those items studied in a given context (i.e., the context of the most recent list). According to Bower's list-discrimination account of inter-list repetition effects (Anderson & Bower, 1972; Sternberg & Bower, 1974), following the retrieval of an item and prior to its production, subjects match contextual information retrieved by the item against the context specific to the current list. This account predicts that inter-list repetition will impair the recall of repeated items because the retrieved contexts from earlier presentations will not match the context of the target list.

Anderson and Bower's manipulation of inter-list repetition in free recall mirrored related literature investigating PI

in immediate serial recall of three-item lists. In these procedures, subjects studied successive lists featuring items drawn from a single semantic category. After several similar lists, the category switched (Loess, 1967), resulting in a profound "release" from the effects of PI (Craik & Birtwistle, 1971; Wickens, 1970).

To understand the interaction between memories formed in distinct lists, Zaromb et al. (2006) studied delayed free recall of word lists that included mixtures of new items and repeats from specific earlier lists. They hypothesized that recalling a repeated item would evoke the associated contextual cues from both the target list and the earlier list containing the repeated item. Whereas retrieval of the target-list context would lead to the standard contiguity effect (i.e., the tendency to successively recall items studied in neighboring positions) retrieval of the earlier list context would induce subjects to commit prior-list intrusions (PLIs). Consistent with this hypothesis, they found that PLIs occurred more frequently following recall of repeated than novel items.

Although interlist repetition led to an increase in PLIs following recall of repeated items, neurologically healthy subjects rarely commit PLIs in standard free recall or even when

recalling lists with some overlapping items. That subjects rarely commit intrusions in free recall underscores the additional role of source-monitoring in episodic retrieval (for a review see Mitchell & Johnson, 2000). In free recall, subjects may remember non-presented words to the degree that they possess semantic, phonological, or perceptual similarity to current list items (Roediger & McDermott, 1995). Whether such words appear as intrusions depends on subjects' ability to determine their correct source. The interlist repetition procedure can thus help researchers investigate the role of source monitoring in recall tasks by activating items learned in the wrong context and challenging subjects to filter out those responses.

The List-Before-Last Procedure

Whereas the inter-list repetition procedure manipulates PI, the list-before-last (LBL) procedure manipulates RI. Rather than asking subjects to recall the just-presented list in the face of interference from prior lists, the LBL procedure asks subjects to remember an earlier list in the face of interference from a more recent list. Figure 1B illustrates the basic methodology. Subjects study and recall a series of lists; after each list, they attempt to recall items from the "list-before-last" (Jang & Huber, 2008; Lehman & Malmberg, 2009; Sahakyan & Hendricks, 2012; Shiffrin, 1970; Unsworth, Spillers, & Brewer, 2012; Unsworth, Brewer, & Spillers, 2013; Wahlheim & Huff, 2015; Wahlheim, Richmond, Huff, & Dobbins, 2016; Wahlheim, Alexander, & Kane, 2019; Wahlheim, Ball, & Richmond, 2017; Wahlheim & Garlitch, 2020; Ward & Tan, 2004).

In standard free recall, there is minimal RI, as recall follows immediately after the presentation of the study list. As such, it is not surprising that subjects are generally better at recalling items from the most recent list than from the list before last. In everyday life, episodic retrieval rarely follows so closely on the heels of the initial experience. More likely, the experience and the retrieval suffer from interference related to the encoding and retrieval of other memories. Despite this increased interference, subjects typically make an order of magnitude more correct responses from the target list than intrusion responses from the intervening list (Jang & Huber, 2008; Sahakyan & Hendricks, 2012).

Shiffrin (1970) introduced this procedure as an empirical challenge to classic theories which assumed RI reflected a combination of competition between new and old associations and unlearning of old associations caused by the learning of new associations (see Postman & Underwood, 1973, for a review). If studying intervening items causes unlearning of target items, then quadrupling the intervening list length should worsen target-list recall. Shiffrin (1970) found that the length of the intervening list did not affect target-list recall rates (Jang & Huber, 2008; Sahakyan & Hendricks, 2012; Unsworth et al., 2012; Ward & Tan, 2004) and inter-

preted this invariance as strong evidence against unlearning. Rather, Shiffrin suggested that any difficulties subjects experience retrieving LBL items arose from a failure to retrieve the context of the target list.

Rather than causing unlearning of old information, the prevailing evidence suggests that performing recall between the target and the intervening lists protects target-list memory from RI from the intervening list. For example, if a brief pause, rather than a recall period, separates study of the target and intervening lists, then target list recall decreases as the length of the intervening list increases (Jang & Huber, 2008; Ward & Tan, 2004). In a series of experiments that varied the nature of the task subjects performed between lists, Jang and Huber (2008) found that between-list tasks requiring episodic memory retrieval yielded consistent target-list recall levels despite varying lengths of the intervening lists. These results underscore the role of episodic memory retrieval in influencing RI, such that stronger RI effects reflect a loss in accessibility of relevant retrieval cues.

Theoretical accounts of data obtained using this procedure often assume that subjects associate a list-specific context with the target list and that the nature of the intervening material (e.g. intervening list length, recall or pause between lists) influences accessibility to the list context (Jang & Huber, 2008; Lehman & Malmberg, 2009). However, these models leave open the question of how the intervening material influences accessibility. As an attempt to explain these processes, Lohnas, Polyn, and Kahana (2015) developed a model which implements retrieved-context theory of memory (Howard & Kahana, 2002; Sederberg, Howard, & Kahana, 2008; Polyn et al., 2009) with a post-retrieval editing process, similar to that proposed by Anderson and Bower (1972). This model, termed the continuous memory version of the context maintenance and retrieval model, or CMR2, assumes that each studied item is associated with a slowly drifting temporal context state. This context state serves as the recall cue, and retrieving an item leads to retrieval of its associated temporal context from study. In standard free recall, CMR2 only overtly recalls an item retrieved from memory if the retrieved item's associated temporal context is sufficiently similar to the current context state. However, in the LBL procedure, CMR2 filters recalls by requiring a retrieved item's context to not be too similar to the current context at the start of recall, because then this item would most likely be from the intervening list. At the same time, an item's associated context cannot be too dissimilar to the current context, because then the item is most likely from a more distant list.

CMR2 can account for several key findings from the LBL procedure. With a longer intervening list length, temporal context drifts further away from the target list context, and thus it is less likely that CMR2 can use its current context state to recall target list items. As a result, CMR2 predicts reduced target-list recall for longer intervening list lengths

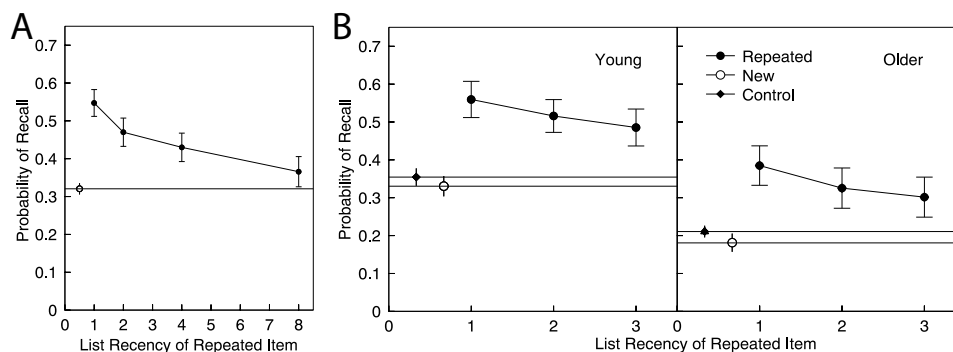


Figure 2. Effects of interlist repetition on free recall. A. When recalling a list of words that includes some words that occurred on prior lists, Zaromb et al. (2006, Experiment 1) found that subjects exhibited better recall for repeated items, especially when they occurred on recent prior lists. The probability of recalling once-presented items is designated by a horizontal line corresponding to an open circle. B. Younger and older adults exhibited similar interlist repetition effects (Zaromb et al., 2006, Experiment 2). Here, each list contained items from 1, 2, and 3 lists back. The probabilities of recalling new items in experimental and control trials are designated by the horizontal lines corresponding to open and darkened circles, respectively. Error bars are 95%-confidence intervals calculated according to the method of Loftus & Masson (1994).

when there is only a pause between lists. However, this dynamic changes when there is a list-before-last recall period between each list. In this case, after the target list ($n - 1$) is presented, but before the intervening list (n) is presented, CMR2 searches for items from an earlier list ($n - 2$). This contextual retrieval clouds the current context state, generally making it more difficult for the model to retrieve target list items during the next recall period. Importantly, this increased difficulty is due to retrieval from the earlier list, rather than the presentation of the intervening list, which allows CMR2 to capture the finding that intervening list length doesn't affect target list recall performance.

However, not all evidence favors CMR2's approach. Sahakyan and Hendricks (2012) posited that, if more items are recalled in the period between the target and intervening list, then this should cloud the context representation further and thus reduce target-list recall. Sahakyan and Hendricks (2012) manipulated the delay between study and test of the list before the target list, and found that delay length reduced the number of recalls from the list preceding the target list. However, in the subsequent recall period, delay length did not influence target list recalls nor intervening list intrusions. Sahakyan and Hendricks (2012) took these results as evidence against CMR2's explanation regarding the importance of context retrieval during each intervening recall test. However, Lohnas et al. (2015) proposed that the search process itself might be sufficient to cloud the current context state, regardless of how many items subjects overtly recall.

The unique nature of the LBL procedure challenges theories that lack a mechanism of list differentiation. Lohnas et al. (2015) solved this problem by assuming that subjects cue recall using the state of context at the end of the intervening list and then use context-similarity to filter out inappropriate

recalls. Healey and Wahlheim (2023) proposed a more direct process of targeting the appropriate list, by assuming that subjects encode then partially reinstate a start-of-list context (similar to Lewandowsky & Murdock, 1989). Direct reinstatement of the target-list context then promotes recall of other target-list items.

Although Healey and Wahlheim (2023) assessed this mechanism only with a single set of list-lengths and a pause between lists, other models assume similar automatic target-list context reinstatement which varies with intervening list length and task between lists (Jang & Huber, 2008; Lehman & Malmberg, 2009). As a more direct investigation of target-list context reinstatement, Unsworth and colleagues tested the account that the target-list context can be isolated and reinstated, by examining recall performance and recall latencies. They found that patterns of performance were most consistent with an account assuming that the reinstated target list contains "noise" from intervening list and prior list items which have shared context features with target-list items (Unsworth et al., 2012, 2013). Building on prior work examining recall probabilities, they found that recall latencies increased in the presence of an intervening list but did not change with intervening list-length. This provides further evidence that intervening list items compete for recall, but their contribution to the competition is independent of intervening list-length when there are recall periods between lists. Taken together, these theoretical accounts of the LBL task enrich our understanding of how the reinstatement of a target list context can be affected by and insulated from both proactive and retroactive interference from other experiences.

The Recall-by-Category (RBC) Procedure

The inter-list repetition and the LBL procedures both tax subjects' abilities to focus retrieval on the target list. In the inter-list repetition procedure, the overlapping content of the lists makes it more difficult to selectively retrieve items from the most recent list. In the LBL procedure, the presentation of the intervening list and recall of earlier lists both interfere with recall of the target list. Here, we consider a procedure that provides subjects with a retrieval cue that directly specifies a subset of items from the target list. In the recall-by-category (RBC) procedure, each list consists of words drawn from a small number of categories. However, rather than allowing the list itself to serve as the retrieval cue, as in traditional free recall procedures, here the category serves as a cue, and subjects must recall only those list items that belong to the cued category.

The RBC procedure can be contrasted with the more standard procedure of categorized free recall (e.g., Pollio et al., 1969; Wingfield, Lindfield, & Kahana, 1998). In categorized free recall, lists comprise N exemplars drawn from each of M categories for a total of $M \times N$ items (e.g., a 20 item-list may consist of five animal names, five types of furniture, five occupations, and five gemstones). The subject is asked to recall all of the items without regard to the category organization. To infer the effects of category structure on memory, researchers examine the degree to which subjects cluster their responses according to the categorical structure of the lists. The RBC procedure deviates from this standard procedure by creating separate mini-retrieval phases, each cued by a specific category name. This can be contrasted with close variants of the procedure, for example, where the category names are provided, but the subject is free to make recall responses without specific regard to those category cues (e.g., Gershberg & Shimamura, 1995; Wingfield et al., 1998; Mathews, 1954). Another variant involves providing the subject with a subset of the categorized study items to refer to as they perform free recall (Lewis, 1971; Slamecka, 1968; Wood, 1969).

In perhaps the most influential study involving the RBC procedure, Tulving and Pearlstone (1966) found a substantial increase in the number of items recalled with RBC compared to free recall (Dong, 1972). They interpreted this to mean that many item memories are not accessible to the subject in their free recall search, but those memories are nonetheless available, given the appropriate retrieval cue (the category name). They also found that the number of items recalled with RBC was inversely related to the number of study items (the set size) associated with the category, suggesting competitive interference between the items targeted by a given category cue (Raaijmakers & Shiffrin, 1980). Smith, D'Agostino, and Reid (1970) found that performance (in terms of the number of items recalled per category) steadily decreases across a series of RBC periods, providing a clear

example of the phenomenon of output interference in free recall (Dong, 1972; Nickerson, 1984; Roediger & Schmidt, 1980; Smith, 1971).

The preceding examples focus on tasks using taxonomic categories to cue memory search, but the RBC procedure can be used with any feature of the studied materials that defines a specific subset of the studied items. For example, Earhard (1967) used the word's first letter to define the category set, varying the number of study items beginning with a given letter, replicating the set-size interference effect described above. In a particularly clever variant of RBC, Watkins and Peynircioğlu (1983) had subjects study items drawn from three very distinct task contexts, such that neighboring study items were associated with different tasks. Three RBC recall periods followed the list, each targeting one of the three task contexts. Watkins and Peynircioğlu (1983) found that in the later recall periods, even though subjects had already engaged in recall of items from one of the other contexts (presumably a quite distracting mental operation), there was a robust and persistent recency effect. This result was surprising, as studies of the delayed free recall (DFR) paradigm (Postman & Phillips, 1965; Glanzer & Cunitz, 1966) suggested that the first recall period should have disrupted the recency effect for the later recall periods.

Polyn, Erlikhman, and Kahana (2011) used the RBC procedure to examine how a strong semantic cue (a category label) interacts with the temporal structure of a study experience. Items from three categories were interspersed on study lists, such that same-category items never appeared in adjacent serial positions. Like Watkins and Peynircioğlu (1983), they found a recency effect that persists across multiple retrieval periods (Figure 3a). Furthermore, within each recall period, they found temporal contiguity effects spanning the non-adjacent study positions occupied by same-category items (Figure 3b). This suggests that the strong semantic relations among the categorized words allowed the memory system to bridge the temporal gaps separating the categorized words from one another, resulting in both a persistent recency effect and a long-range contiguity effect. CMR proposes that semantic and temporal information are combined during memory search to support retrieval of specific memories (Morton & Polyn, 2016; Polyn et al., 2009). This combination should allow CMR to capture these RBC phenomena, though the model has never been used to directly simulate these data.

Categorized free recall and the RBC procedure have also proven useful in the domain of clinical neuropsychology. (Incisa Della Rocchetta & Milner, 1993) showed that a set of patients with damage to their left frontal lobe were impaired (relative to matched comparison subjects) on categorized free recall, but that this impairment was not apparent when memory was tested using the RBC procedure. They interpreted this as supporting the view that frontal cortex is im-

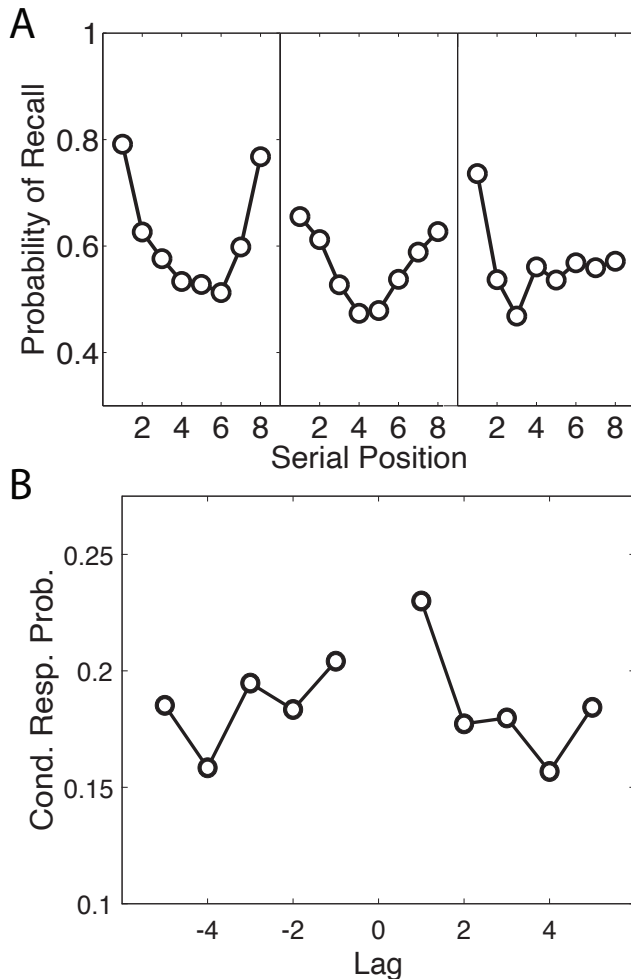


Figure 3. Recency, primacy and contiguity in the Recall-by-Category procedure. A. Within-category serial position curves for the three recall-by-category trials reported by Polyn et al. (2011). From left to right, each panel depicts the probability of recalling the eight studied items belonging to the cued category from the first, second, and third recall period, respectively. B. Lag-CRP analysis averaged across the three recall-by-category periods, showing a temporal contiguity effect for within-category recall transitions, despite those items being separated by multiple other-category items on the original study list.

portant for the strategic use of memory (Moscovitch, 1992), as the provision of category cues in the RBC procedure does some of the strategic work that would otherwise be done by the frontal system (Jetter, Poser, Freeman, & Markowitsch, 1986).

The RBC procedure allows an experimenter to more directly influence the retrieval processes that focus search on particular memories. The category cue can be used to isolate sets of items from interference from the broader list, and

to demonstrate the competitive interference between items from the same category. This procedure also reveals a persistent recency effect that seems more durable than the one observed in standard immediate free recall tasks. Finally, using an RBC category cue reveals that temporal contiguity effects span more than just immediately adjacent study items. The next section expands upon this theoretical point.

The Final Free Recall Procedure

Each procedure discussed above deviates from traditional free recall in manipulating how people search their memories for the target items. The inter-list repetition procedure creates overlapping information across distinct lists, making it harder to focus retrieval on the most recent list. The list-before-last procedure forces subjects to target items from an earlier list but without any explicit categorical or semantic cue to aid their search. In the recall-by-category procedure, each list mixes items from distinct categories, but at recall, subjects must focus their search on just those items that belong to the cued category. Whereas the inter-list repetition procedure requires subjects to use contextual information to overcome semantic interference, the recall-by-category procedure requires subjects to use semantic information to overcome contextual interference.

This section considers the final free recall (FFR) procedure, which expands the timescale of episodic targeting considered with the other procedures. Here, after subjects study and recall many distinct lists over the course of a multi-list session, the experimenter asks them to freely recall as many items as they can remember from across all of the previously studied lists. In FFR, subjects freely search their memories without regard to list boundaries, and they free recall a surprisingly large number of items, further demonstrating excellent retention of information from list to list, even when the final test is given as a surprise (Craik, 1970; Cohen, 1970; Craik, Gardiner, & Watkins, 1970; Dalezman, 1976; Darley & Murdock, 1971; Howard, Youker, & Venkatadass, 2008; Kuhn, Lohnas, & Kahana, 2018; McCabe & Madigan, 1971; Unsworth, 2008). An important feature of this procedure is that the final test is not a pure measure of memory for the initial encoding event; rather, it is a measure of memory for both the initial encoding event and the initial test event, which may or may not have resulted in recall of the target item.

Figure 4 illustrates data from an FFR experiment in which subjects first studied and freely-recalled 16 lists (each comprising 16 common words) and then attempted final free recall of all 256 words (Kuhn et al., 2018). Figure 4 shows the probability of final recall as a function of recency of encoding, considering both the list that each item came from and the item's serial position within that list, for a total of 256 (16 × 16) possible positions. Because retrieval of an item can produce substantial learning, and because spacing between

items affects long-term retention, Kuhn et al partitioned the recall data into three classes of items: Those initially recalled in early output positions (first half of recalls), those initially recalled in late output position (second half of recalls), and those not initially recalled.

Figure 4 illustrates three major effects: First, subjects recalled more items from recent than from remote lists. Second, subjects rarely recalled items that they failed to initially recall. Third, subjects exhibit a pronounced negative recency effect within each list primarily for those items initially recalled in early output positions. Recalling items in later output positions attenuates the negative recency effect.

In a classic FFR experiment, Craik (1970) reported a negative recency effect similar to the one illustrated in Figure 4. He considered two possible explanations: 1) Subjects failed to recall final list items because those items received fewer rehearsals while in short-term memory and thus strong long-term memory representations of those items needed to support final free recall performance were not formed. 2) Subjects failed to recall final list items because those items would tend to have shorter study-test lags during initial recall and thus would not benefit from the well-known spacing effect wherein subjects exhibit superior retention of repeated items when they are widely spaced (e.g., Melton, 1970; Siegel & Kahana, 2014). A series of papers provided support for Craik's first explanation (Craik et al., 1970; Dalezman, 1976; McCabe & Madigan, 1971), leading researchers to disregard and ultimately forget Craik's second explanation. However, Kuhn et al. (2018) presented several data points supporting the spacing account over the rehearsal account of negative recency. They showed that the probability of final free recall increased monotonically with the spacing between the study of an item and its initial recall. Because subjects tend to recall end-of-list items early in output, these items will benefit less from spacing than earlier list items, and those end-of-list items recalled early will have the shortest spacings of all, and thus the lowest recall levels, as seen in Figure 4. Kuhn et al. further examined final recall following lists in which subjects performed a continual distractor free recall task. According to the rehearsal-based account, negative recency arises because final-list items receive fewer rehearsals than midlist items, and as such, the recency items have weaker associative representations in long-term memory. As CDFR should disrupt rehearsal they predicted substantially reduced negative recency in this condition. Contrary to the rehearsal account and in line with the spacing account, Kuhn et al. found strong negative recency in the CDFR condition.

Here we consider another facet of the FFR data; namely, the ability to examine transitions during final recall not only among items within a list but also among items studied on distinct lists. Whereas it is not surprising that FFR transitions between words studied in the same list show a clear contiguity effect (Howard et al., 2008; Unsworth, 2008; Loaiza

& McCabe, 2012), one may ask whether contiguity effects extend across list boundaries when subjects make transitions among items studied on separate lists. In particular, if subjects store unique list contexts for each list, any two lists may be equally similar to each other. Further, subjects may cluster their recalls by list without benefit to the shared temporal information among neighboring lists. By contrast, if subjects associate temporal contexts with each list, as CMR2 assumes, then subjects should be more likely to transition between items studied in neighboring lists.

Figure 5 illustrates the probability that subjects transition, in final recall, among items studied on different lists as a function of the *lag* between the lists (a lag of +1 indicates a transition between an item studied on List *i* and an item studied on List *i*+1). This *across-list contiguity analysis* considers only transitions between correct recalls from one list to correct recalls of a different list and it controls for the fact that it becomes less likely to transition to a particular list as the number of items correctly recalled from that list increases. Following Howard et al. (2008) we also created a surrogate (shuffled) data set to control for the effects of recency and autocorrelated goodness of encoding (each of which could generate a greater tendency for neighboring items to be recalled successively). These surrogate data (gray line in Figure 5) exhibit a substantially smaller contiguity effect than seen in the non-shuffled data.

We have seen how studies of final free recall have provided useful data for understanding the effects of rehearsal, spacing, and contiguity on subsequent recall. But, at a broader level, final free recall provides valuable information on long-term retention of memoranda experienced under short-term memory conditions. Such data require that theories preserve information across multiple interfering lists. Further, by asking subjects to freely recall a large pool of items, one can observe retrieval dynamics under conditions where subjects recall dozens of items. In a multi-session experiment in which subjects, at the start of each session, freely recalled all words learned across all prior sessions, Katerman, Li, Pazdera, Keane, and Kahana (2022) observed average recall rates exceeding 100 words on the final experimental session.

The Overt-Rehearsal Procedure

In this and the next section, we consider procedural variants of free recall aimed at unmasking the covert rehearsal processes hypothesized to occur during learning and the covert retrievals we may filter out during recall. In the overt-rehearsal procedure, Rundus and Atkinson (1970) asked subjects to say, out loud, everything that came into their minds as they tried to memorize the words on the study list for a free recall test. Rundus and Atkinson then counted the number of overt rehearsals each item in the list received. They found that early list items received the greatest number of rehearsals

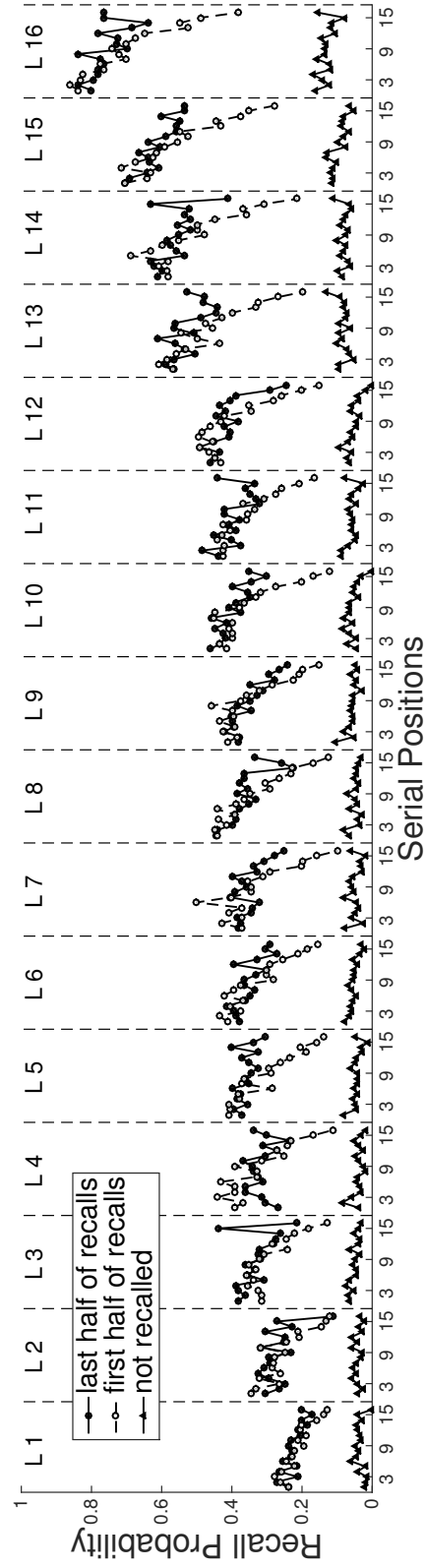


Figure 4. Final-free recall as a function of encoding. We consider three classes of items: Those recalled in early output positions in immediate-free recall, those recalled in late output positions, and items that were not recalled initially but were recalled during final-free recall. We observe a positive long-term recency effect across lists, and a negative-recency effect within lists. The negative-recency effect appears greatest for the items recalled in early output positions. Data from Kuhn, Lohnas and Kahana (2017).

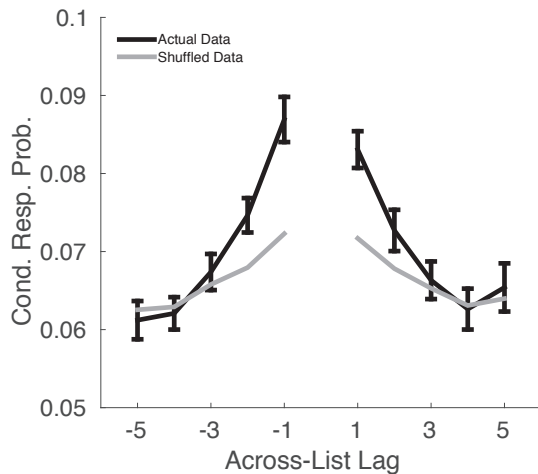


Figure 5. Across-list transitions during final free recall show a contiguity effect. The black curve shows the conditional response probabilities computed from the actual data. The gray curve shows the conditional probability from surrogate data produced by randomly shuffling the order of recalls. Adapted from Healey, Long, and Kahana (2019).

and that subjects often continued to rehearse these items until the end of the list. Further, the number of rehearsals an item received predicted its eventual recall. This finding, coupled with subjects' preferential rehearsal of early list items, supported theories attributing the primacy effect to subjects' rehearsal strategies (Atkinson & Shiffrin, 1968; Raaijmakers & Shiffrin, 1980).

Using overt-rehearsal, Brodie and Murdock (1977) examined the relation between presentation rate and the primacy effect. Previous work has shown that slower presentation rates produce stronger primacy effects. Brodie and Murdock showed that with a slow (5-second) presentation rate, early list items also tend to be rehearsed later in the study list than with a fast (1.25-second) presentation rate. They argued that by rehearsing early list items later in the list, those items benefit from greater *recency*. Thus, the rehearsal process not only leads subjects to devote more time to some items than to others, it also makes items from earlier list positions appear later in the sequence of rehearsals and thus more recent at the time of test.

Brodie and Murdock proposed that rather than examining recall as a function of the position of an item in the list, one should examine recall as a function of the position to which an item was last rehearsed. They thus contrasted nominal and functional serial positions. *Nominal serial position* refers to the position of the item in the list, as presented by the experimenter to the subject. *Functional serial position* refers to the position of the item in the subject's sequence of rehearsals. Because a given item may be rehearsed many times during the study period, Brodie and Murdock decided

that the most relevant position is that of the last rehearsal. Whereas Brodie and Murdock found pronounced primacy in the nominal serial-position curve, especially in their slow presentation rate condition, primacy vanished in their analysis of functional serial-position curves. Rather, the functional serial position curve showed a nearly continuous recency effect extending from the end of the list to the beginning. This analysis suggests that the primacy effect largely results from early list items being rehearsed more recently than middle items. Subsequent work by Ward and colleagues (e.g., Tan & Ward, 2000; Ward, 2002) replicated and extended these findings by showing that in a broad range of experimental conditions, the recency with which an item has been last rehearsed is the critical variable predicting its recall.

Murdock and Metcalfe (1978) examined the distribution of rehearsals following the presentation of each study item. They found that subjects tend to rehearse recent items and items from the start of the list more than other items. The rehearsal mechanism thus generates a primacy and recency effect resembling those seen in free recall itself. This and other recall-like rehearsal phenomena led Laming to argue that rehearsals are essentially mini-recalls that subjects make as they study the list (2006; 2008). Laming's view turns the analysis of rehearsals on its head. Rather than using rehearsals to explain recall, Laming argues that we should use recall to explain rehearsal. In free recall, subjects do not merely attend to the presented stimuli; rather, they use the study period to think about (i.e., recall) previously studied items. Data generated by the overt rehearsal procedure thus cannot establish a causal linkage between rehearsal and recall phenomena.

Though treated as separate stages of a memory task, encoding and retrieval occur in tandem. Elaboration during encoding entails retrieving associated representations from memory, and subjects often deliberately "rehearse" earlier list items as an encoding strategy. The overt rehearsal procedure makes such retrievals explicit by asking subjects to articulate every item that comes to mind during encoding. Even though such "forced" retrieval likely differs from natural retrieval processes during free encoding, overt rehearsals exhibit all of the usual properties of standard free recall, such as recency, primacy, and contiguity (Bhatarah, Ward, & Tan, 2006).

Critics of the overt-rehearsal procedure have suggested that any attempt to explain recall data with rehearsal data suffers from the chicken-and-egg problem. Unless we know why subjects rehearse items the way they do, we can't use their rehearsal to explain their recall. Instead, their rehearsal data may only provide us with one more thing that requires explanation—namely, what makes rehearsals? One way around this problem is to instruct subjects to rehearse in a particular manner (Tan & Ward, 2000; Murdock & Metcalfe, 1978), but exerting control in this way may substantially alter

the nature of the rehearsals themselves (see Ward, in press, for a discussion of this point).

The Externalized Free Recall Procedure

Ever since Melton and Irwin (1940), students of memory have recognized that failures of memory search appear not only as omitted responses but also as intrusions of items from outside of the target list. Deese (1959) and Roediger and McDermott (1995) demonstrated how semantic associations cause subjects to commit intrusions in free recall, often with high confidence. Zaromb et al. (2006) found a similar effect for temporally-defined associations. Following recall of a target list item that appeared on a prior list, subjects exhibited a marked increase in their tendency to make intrusions, often from the prior list on which the repeated item appeared (see section "The Inter-List Repetition Procedure"). However, under most circumstances, intrusions occur infrequently as subjects likely censor themselves, omitting responses they deem inappropriate. Thus, the small number of *overt* intrusions that people commit during recall (typically around 5% of the total number of recalls) likely underestimates the role of intrusions in the recall process.

To study the influence of covert intrusions on recall, Kahana et al. (2005) introduced an externalized free recall (EFR) procedure adapted from an earlier method developed by Roediger and Payne (1985) and Bousfield and Rosner (1970). In the EFR procedure, after subjects have become familiar with standard free-recall instructions, the experimenter asks them to say out loud all words that come to mind at the time of test, even if they think those words did not occur in the most recent target list. To separately examine the internal censoring process during recall, researchers using the EFR procedure may also ask subjects to indicate when they have recalled an item that they believe was not on the most recent list by pressing a key immediately following recall of that item (in the Roediger and Payne method, subjects were asked to recall every word that came to mind, but they were not instructed to indicate which responses were errors).

Some theories assume that subjects generate more items than they actually recall but apply an internal filter to only report items as recalled from the current list. If this assumption is valid, the EFR procedure asks subjects to recall *all* generated items while making their filtering criteria explicit (by rejecting responses deemed inappropriate to the target context). Lohnas et al (2015) used EFR to evaluate the role of generate-recognize processes in their retrieved-context model of memory search (CMR2). Because CMR2 uses a context-comparison criterion to determine whether to recall a retrieved item, it is straightforward to simulate EFR data by allowing the model to recall all of the retrieved items and then having the model reject any item that fails the context-comparison criterion. CMR2 assumes that both accepted and rejected retrievals update context.

Comparing EFR and standard FR in a large dataset, Lohnas et al. (2015) demonstrated that EFR does not fundamentally change the processes that govern free recall (and context updating). Although subjects made many more intrusions in EFR, recall of correct items did not differ reliably between groups, suggesting that the increased recall of intrusions in EFR did not hinder the recall of correct items. In addition, the recency advantage in recall initiation and recall probability was preserved between the two tasks. Lastly, properties of recalled intrusions were preserved as well: Subjects recalled as many PLIs in immediate free recall as the number of nonrejected PLIs in EFR. Lohnas et al (2015) also showed that the same parameterization of the CMR2 model matched data from both EFR and standard free recall. These results buttress prior studies demonstrating that EFR instructions do not significantly interfere with the normal recall process (Kahana et al., 2005; Unsworth & Brewer, 2010; Zaromb et al., 2006).

CMR2 rejects retrieved items when the context they retrieve does not adequately resemble the target context, which in the case of immediate recall is the context at the time of test. Consequently, the model predicts higher rejection probability for PLIs than for correct items and as a higher rejection probability of PLIs from more distant than recent lists (Unsworth, Brewer, & Spillers, 2010). Lohnas et al's EFR study verified these predictions, supporting the idea that subjects use internal context states to filter out retrieved items.

To the extent that EFR provides a window into subjects unfiltered thoughts during free recall, it enables a much richer analysis of recall dynamics leading to errors and even the way errors cue other memories. Consider data from an experiment where items repeat across lists (e.g., the inter-list repetition procedure). Correct recall of a repeated item should elicit retrieval of items from both of its associated lists, even though items from its first, prior list are errors. Zaromb et al. (2006) examined inter-list repetitions with an EFR procedure and found that when two PLIs occur in succession, and when they both came from the same list, they tend to come from neighboring list positions (Figure 6). Similarly, in lists without repetitions, Unsworth et al. (2010) found that a PLI was less likely to be marked as an intrusion when it followed immediately after a correct response, as compared to PLIs that followed immediately after another PLI. They interpreted this finding as evidence that PLIs with a stronger connection to the target list context are more likely to be misinterpreted as correct responses. Unsworth et al. (2010) also found that rejection probabilities of intrusions were lower for early than late output positions, again suggestive of the stronger associations with the correct-list context used to initiate recall.

The EFR procedure has also clarified mechanisms underlying group differences in overall recall performance. In standard free recall, healthy younger adults with low working

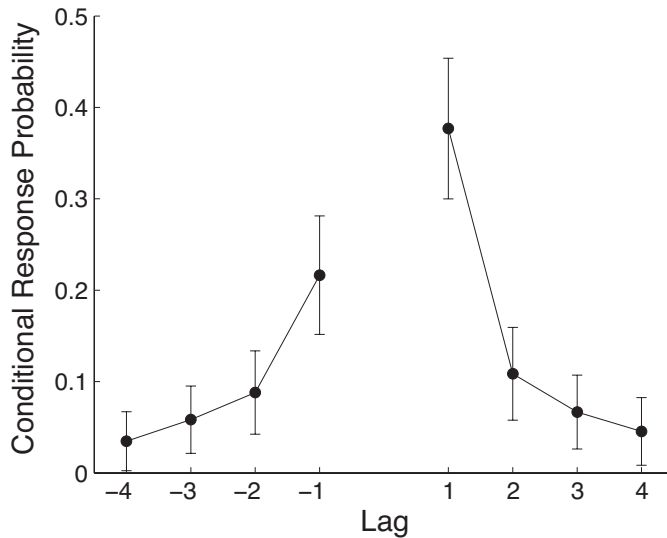


Figure 6. Temporal contiguity and prior-list intrusions (PLIs). Successive PLIs that came from the same original list tend also to come from neighboring serial positions. Based on studies in the free-recall database (Sederberg et al., 2010) and a previous analysis reported in Zaromb et al. (2006).

memory capacity recall fewer correct items and make more intrusions compared to younger adults with higher working memory capacity (Unsworth, 2007; Unsworth et al., 2010). The EFR procedure revealed that the increase in intrusions reflects reduced rejection probability and greater overt generation of PLIs. Similarly, older adults also recall fewer correct items and commit more intrusions than young adults in standard free recall (Kahana, Howard, Zaromb, & Wingfield, 2002; Rhodes, Greene, & Naveh-Benjamin, 2019). As with low-capacity young adults, EFR reveals that older adults both generate more intrusions and exhibit lower rejection probabilities than younger adults (Kahana et al., 2005; Wahlheim et al., 2017).

The spatial-episodic memory procedure

Episodic memories unfold in both time and space (Tulving, 2002; Kahana & Miller, 2013), yet free recall procedures rarely manipulate the spatial context of memoranda. To study the joint contributions of temporal and spatial information to memory search, Miller, Lazarus, Polyn, and Kahana (2013) developed a hybrid spatial navigation-verbal free recall task. In their *Courier* procedure, subjects play the role of a courier in a 3D-rendered virtual town (Fig. 7). After an initial town familiarization phase, subjects begin a series of “delivery days,” each of which involves experiencing a series of objects at different spatial locations and then freely recalling those objects. The encoding phase of each “list” takes place during gameplay.

On each delivery day, subjects navigate to a series of randomly chosen list-unique stores, chosen without regard to the stores’ spatial proximity to one another (this was done to decouple the effects of spatial and temporal proximity during encoding). Upon arrival at each store, the game informs subjects of the identity of an object they delivered. At the final store on the delivery route, the screen goes blank and subjects attempt to freely recall the delivered objects from that delivery day. The *Courier* task departs from typical free recall methods in the encoding phase of the task, where each memorandum occurs in a (virtual) spatial context and where subjects use their spatial memory for navigating to the target store between each delivery, thus potentially disrupting the strategies they might otherwise use to associate neighboring items (e.g., rehearsal). The interpolated store-navigation between item deliveries, which typically takes 15-60 seconds makes the task resemble continual-distractor free recall, where subjects perform an interpolated distractor task between item presentations.

By exploiting the technology of desktop virtual reality, *Courier* allows researchers to probe memory in a quasi-naturalistic setting while still providing precise experimental control. Whereas daily life generally does not involve learning a random list of words presented in rapid succession, *Courier* embeds these items in a spatial layout that does not change over trials, thus creating a contextual scaffold for the otherwise unrelated memories. Investigating recall in *Courier* allowed Miller et al. (2013) to demonstrate spatial clustering of delivered objects. Although spatial information had previously been studied in verbal and visual episodic memory using words or objects presented in 2D arrays (e.g., Mandler, Seegmiller, & Day, 1977; Rothkopf, 1971; Naveh-Benjamin, 1987; Cortis, Dent, Kennett, & Ward, 2015) Miller et al demonstrated spatial organization of memories experienced during first-person navigation. Much as conditional-response probability curves decrease as the temporal lag between memoranda increases, in this task the tendency to cluster items decreased as the spatial distance between those items increased (see Figure 7). More recently, Dougherty et al. (in press) have shown that spatial clustering occurs based on Euclidean distance, providing a link to cognitive map theories of memory (Herweg & Kahana, 2018).

In combining spatial and verbal memory in a single task, *Courier* also helps to bridge studies of the neural bases of human and animal memory (Herweg & Kahana, 2018). Whereas decades of neuroscientific research have described how animals build representations of space and use those representations to support goal-directed learning (O’Keefe & Nadel, 1978), most human cognitive neuroscience experiments involve sequences of memoranda – often verbally labeled – that unfold in time. This wide gap between methods makes it difficult to build a common base of knowledge on commonalities across species. Miller, Neufang, et al. (2013)

had neurosurgical patients fitted with hippocampal depth electrodes perform the Courier task. They sought to evaluate whether neural codes representing a spatial map form part of the spatiotemporal context hypothesized to underlie human episodic memory. Using specialized research electrodes that recorded both field potentials from larger contacts and individual neurons from smaller contacts, they could measure the activity of hippocampal neurons during gameplay and relate this activity to the subject's location within the virtual environment. These analyses revealed that a significant fraction of hippocampal neurons responded selectively when subjects navigated through particular regions of the virtual town. Researchers have extensively documented and characterized the behavior of such place-selective neurons, or *place cells*, first discovered in rodents (O'Keefe & Dostrovsky, 1971) but also seen in the human hippocampus during virtual spatial navigation (Ekstrom et al., 2003; Jacobs, Kahana, Ekstrom, Molison, & Fried, 2010).

Miller et al (2013) identified place cells (comprising 24% of all observed neurons) during the navigation phase of the Courier task. Hypothesizing that these cells represent the spatial context of the items delivered at specific locations, they asked whether the ensemble neural activity associated with these places reinstates in the moments preceding the free recall of a delivered item. Indeed, they found that during the spontaneous recall of an item, place-responsive neurons exhibited firing patterns similar to the patterns they exhibited during navigation through the region of the town where the item was delivered (see Figure 7). Miller et al.'s finding that spontaneous recall of an item reactivates its spatial context provides neural evidence for theories of episodic memory that postulate context reinstatement as the basis for recollection (Kahana, 2020). This finding further suggests that the spatial coding identified with the hippocampal place cell system is part of a more general engine of episodic memory in which items become associated with their spatiotemporal contexts and retrieval of items reinstates those contexts to help cue other context-appropriate memories. Recent findings of time cells in the human brain have similarly been linked to the contiguity effect in traditional verbal free recall (Umbach et al., 2020). In related work, Herweg et al (2020) investigated the role of hippocampal theta oscillations in memory retrieval by having neurosurgical patients fitted with hippocampal depth electrodes play Courier. They found that increases in 4-8 Hz hippocampal theta activity predicted the degree to which retrieved memories exhibited spatial clustering.

Although students of memory typically assume that each item in a word list has a unique context, the contextual diversity within word list experiments is impoverished and contrived. The semantic structure of a word list determines the dimensions along which context varies in such experiments. For experiments that randomly choose words from

a large lexicon, the semantic context for each word jumps around haphazardly through the list. For experiments that draw items from one or more semantic categories, the semantic context is relatively constant or jumps suddenly after a change in category. In contrast, the Courier procedure creates a continuously varying spatial context for each encoded item. It also enhances the diversity of temporal context by introducing a secondary task (navigation) between each delivered item. Because efficient navigation between stores requires mental effort, this secondary task should reduce the degree of inter-item rehearsal, much like a continual-distractor free recall procedure (Bjork & Whitten, 1974; Howard & Kahana, 1999). Although Courier departs in many ways from real-life experience, many subjects find the gameplay enjoyable and the cover story compelling.

Despite the many differences between Courier and standard word list free recall, Courier exhibits the usual behavioral effects that characterize word list recall, including primacy, recency, contiguity, and similarity (Miller, Neufang, et al., 2013; Dougherty et al., in press). In addition, Courier gives rise to reliable levels of spatial clustering, even when subjects have no specific reason to recall the spatial information of the studied items. Finally, Courier generally gives rise to fairly high levels of recall, possibly due to the rich contextual information surrounding each item.

The courier task is not without limitations. One particular weakness concerns the slow rate of data acquisition. Because subjects take an average of ~30 seconds to navigate from store to store, studying and recalling a list of 16 items will require nearly 10 minutes. This results in a considerably slower data acquisition rate than standard word list recall tasks, but not much slower than continual-distractor recall tasks. Another limitation of Courier common to all VR-based tasks is that approximately 10% of subjects will experience nausea during gameplay. Nonetheless, Courier's ability to generate a rich spatial context for studied items, high levels of recall, and neural correlates of time and space make it an appealing addition to the memory researchers' toolbox.

Summary and Synthesis

Each of the seven specialized recall procedures reviewed here expands the boundaries of traditional recall paradigms in various ways. While standard free recall uses a unique set of items on each trial, the inter-list repetition procedure challenges memory by including, within each list, a subset of items learned on prior lists. By eroding the contextual boundaries between lists, this procedure challenges subjects to focus their retrieval on the target list while inhibiting neighbors of the repeated items experienced on earlier lists.

The list-before-last procedure challenges memory in a different way. Whereas standard free recall asks subjects to retrieve the set of most recently experienced items, the list-before-last procedure asks subjects to avoid recalling those

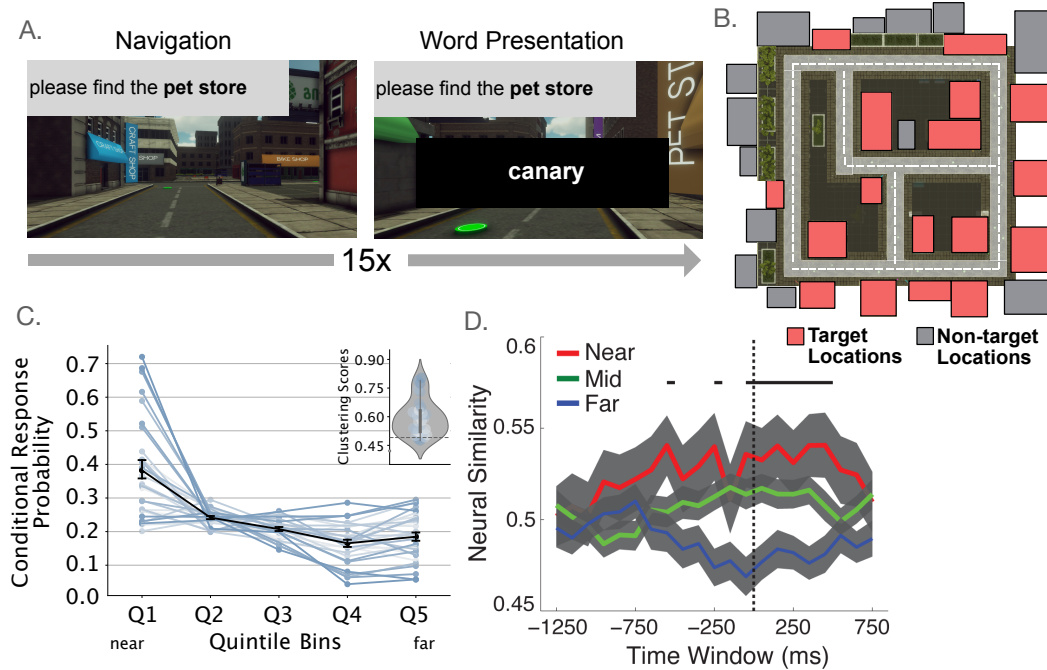


Figure 7. The Courier Task. A. Subjects navigate along a courier’s delivery route in a virtual town, delivering an object to each business they visit (e.g., delivering the canary to the pet store). Subjects subsequently recall each of the delivered objects, either freely or in response to cues. B. Overhead view of the town. C. Spatial clustering appears as an increased tendency to make transitions during recall to objects delivered in nearby spatial locations (Dougherty et al., in press). D. Data from Miller et al (2013b) showing the average neural similarity for near, middle, and far spatial distance bins between ensemble place-responsive cell activity during navigation and place-responsive cell activity during item recall as a function of time relative to recall onset, computed in overlapping 500-ms time windows (x-axis values indicate the center of the time window). Shaded regions indicate *SEM* across recalled items. A horizontal bar indicates statistically significant time points.

prepotent items and instead focus their memory search on those items experienced on the preceding list. Because these lists typically draw items from a common pool of words, temporal context is the only cue that can be used to disambiguate the lists in memory.

The recall-by-category procedure changes the targeting from an entire list to a subset of items that share semantic features. This matches many situations in daily life where you recall, e.g., all of the vegetables you bought on a recent shopping trip, or all of the interactions you had with a particular coworker. Recall by category allows researchers to manipulate the targeting process that occurs during memory search. Doing so has led to insights into the nature of within-list interference, and has provided evidence for the persistence of recency and contiguity across multiple recall periods.

Rather than narrowing the set of contextually appropriate responses, as in the recall-by-category procedure, the final-free-recall procedure broadens the set to include all lists experienced by the subject. Here, subjects attempt to recall items irrespective of list membership as long as they occurred during the more general context of the experiment. Subjects

in final free recall typically recall a much larger number of items but a smaller fraction of the items recalled on any given list. If one conducts final free recall across multiple days of an experiment that uses a single large word pool, subjects can freely recall more than 100 unique words in a single recall period (Katterman et al., 2022). Final free recall reveals how recency influences recall at short and long time scales and how contiguity spans many items and multiple lists (Figure 5).

The overt-rehearsal procedure aims to shed light on the strategic processes occurring during intentional memory encoding. Subjects frequently report verbally rehearsing words, and such rehearsals would be recorded in memory and shape subsequent recall. The overt-rehearsal procedure specifically asks subjects to verbalize these rehearsals and shows that the frequency and temporal pattern of vocalized rehearsals predict important features of recall. Data produced in overt-rehearsal experiments has helped memory scientists address various theoretical questions about memory, as summarized in a review by Ward (2022).

The externalized free recall procedure also changes the contextual targeting of memories. But rather than manipu-

lating the structure of the lists, this procedure asks subjects to recall any item that comes to mind rather than to inhibit contextually inappropriate responses. The procedure further asks subjects to report when they recognize a retrieved item as coming from the wrong context. Externalized recall thus reveals information about memory search that would be obscured when subjects know that they must avoid recalling items that do not belong to the target list.

In the spatial-episodic memory procedure, subjects experience items within a spatial context. In this free-recall variant, subjects play a game in which they learn the layout of a computer-generated virtual town as they navigate it from a first-person perspective, playing the role of a courier delivering objects to businesses. This game exploits modern computing technology to break out of the traditional discrete item presentation format that has characterized memory research since the creation of the memory drum at the turn of the 20th century. Using a quasi-naturalistic format of item presentation, the Courier game creates both a spatial context for each memorandum and a natural separation between memoranda governed not by the timing of the experiment but by the subject's navigational behavior. The dynamics of recall in this procedure reveal the joint influence of temporal and spatial context on memory retrieval. The quasi-naturalistic nature of encoding, in which a complex navigational task separates the encoding of items, reveals the degree to which principles of memory search generalize beyond the setting of traditional word list experiments.

Standard free recall, serial recall, and paired associate procedures (we only focused on free recall) have generated valuable data for over a century and will continue to have currency in our quest to understand memory at a psychological, biological, and even societal level. The standard versions of these procedures arose from a particular set of historical pressures. Starting around 1960, short-term memory procedures, such as immediate free and serial recall, rapidly generated data showing how variables such as list position, modality, presentation rate, repetition, and spacing influence recall performance. Theories emerged to explain existing data and generate predictions to be verified in further experimental studies (Murdock, 1974). Laboratory computers that replaced slide projectors and memory drums (in the 1970s) provided even greater experimental control. It is not an accident that these classic procedures survived and thrived. At the same time, researchers have long recognized limitations of these standard procedures which can be overcome by methodological innovations. The present review considered several procedural variants that aim to overcome specific limitations of traditional free recall, as described above.

We see particular value in studying people's ability to target memories outside of the context of a recently presented list and in the presence of overlapping cues. Memory is not a tabula rasa and theories need to tackle the challenging ques-

tion of how subjects focus their search on the most recent list while filtering out memories from prior lists (Lohnas et al., 2015). Further, we see the analysis of recall errors as providing particularly important insights into the process of contextual targeting as seen in intrusions coming from prior lists (Zaromb et al., 2006) and intervening lists (Jang & Huber, 2008). To assess intrusions, it is helpful to encourage subjects to speak any item that comes to mind and mark those items they deem intrusions, as in the externalized recall procedure. Finally, we see modern computing technology as offering us the ability to advance our understanding of memory by embedding the demands of a traditional memory experiment within a more naturalistic situation, thus helping to ensure that the memory mechanisms uncovered in the laboratory generalize to people's everyday lives (Dougherty et al., in press).

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